connected over the network as well.

We have tested it with two different types of cores as well, namely, OpenAirInterface (OAI) 5G core and the Open 5GS core. These cores run on different systems.

The USRPs (refer Figure 1) act as gNB base stations and can be connected to either user's equipment software. It could connect to another USRP and also to another 5G phone.

Experiments performed so far

We have conducted multiple experiments that include hands-on experiments with the RAN. We have looked at the impact of different parameter settings. We have also looked at different types of real and simulated systems with different types of antennas and how that affects the connectivity and the data rates thus procured.

Core network allows us to analyse various entities and understand the configuration of AMF, SMF, UPF, etc. It also allows us to verify the sequence of events logged at gNodeB and CN. It provides us with a better understanding of SIM card functions.

TSDSI interaction and work in progress

The vision of TSDSI is to ensure that digital communication standards increasingly drive domestic, economic, and policy activities and enhance India's competitiveness for ICT goods and services in global markets. It aims to do this by creating a leadership position through India's participation and contribution to emerging digital communication standards in global SDOs.

Recently, TSDSI SGI Standard & NaviC (Satellite based Indian Navigation System) have become part of 3GPP. The CPRI Fronthaul Transport standard developed by TSDSI has been taken up by TEC for adoption as a National Standard. Open source begins where standardisation efforts end. They both work hand in hand and can not only co-exist but also augment each other.

5G Hackathon India

We contributed to the 5G Hackathon India where our three entries were selected among top 100 out of about 1500 entries:

- Physical Layer Enhancements for 5G in the Indian Context (Selected among TOP 30)
- Advanced MM-wAve Systems for INformatics at Gigabit (AMMAZING) (Selected among TOP 30)
- Robust Tele-Driving over a 5G Network (Selected among the TOP 100)

TSDSI role in open source

TSDSI has also established an open source task force to study and evaluate open source components required for building an end-to-end system. It explored how open source projects can accelerate the Indian 5G ecosystem. Weekly meetings were conducted for over nine months with broad industry as well as academic participation. The findings were published in August 2021. Some potential impacts are:

- It is a game changer for original equipment manufacturers (OEMs) and service providers.
- Expedites SGI for rural India.
- It reduces the cost of implementation by about 50-60%.
- Comprehensive, well tested open source platform with support.

Building such an open source platform, which can actually reduce the development cost for OEMs, allows project developers to focus on their core competence. We want to keep our approach similar to that of the Red Hat model. There will be basic software available for free and you will have to pay in order to avail more services.

The overall idea is to have three pillars on which this will be built:

- Technical leadership from academia with well-experienced faculty from academia who have experience with several successful projects.
- Dedicated core development team consisting of a team with prior industry experience.
- A strong industry participation through volunteerism to take on specific sub projects and projects of interest.

IOS - 5G scope

So, in simple terms, there are just two deliverables which we are aiming at. The first one is the DU/CU and the second is the core 5G part of the network. The second part is going to be the RAM intelligent controller (RIC) and the service management orchestration in a cloud based deployment.

In the end, it is important to note that the 5G testbed is open for people to use. In order to use the testbed you can visit *5Gtestbed.in* where all the currently offered services can be found. One just has to register their company and then they can start using the testbed. END

This article is put together from a tech talk session by Dr Chandra R. Murthy of the Indian Institute of Science at Open Source Conference 2022: Proliferation of Open Source for Emerging Technologies, organised by Samsung R&D Institute India - Bangalore and IEEE ComSoc Bangalore Chapter. It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY with a knack for open source exploration and research.

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The article was originally published in the December 2022 issue of Electronics For You.